BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, AJK, MIRPUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: 022/1

Retyped faithfully from the original board paper — same questions, no changes.

CHEMISTRY (PART –II) (OBJECTIVE PART)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 022/1

Note: Note:- Write your Roll No. in space provided. Over writing, cutting, using of lead pencil will result in loss of marks. All questions are to be attempted.

Q1. (Marks (17)) Each question has four possible answers, Tick (√) the correct answer.

Q1. Mark the correct statement;

(A) Metallic character increases down the group

(B) Metallic character increases from left to right along a period

(C) Metallic character remains the same from left to right along a period

(D) Metallic character remains the same down the group

Q2. Which of the following is not an alkali metal?

(A) Francium

(B) Cesium

(C) Rubidium

(D) Radium

Q3. The chief ore of aluminium is;

(A) H_3BO_3

(B) $Al_2O_3.2H_2O$

(C) Al_2O_3

(D) $Al_2O_3.H_2O$

Q4. Which of the following species has the maximum number of unpaired electrons?

(A) O_2

(B) O_2^+

(C) O_2^-

(D) O_2^{-2}

Q5. Hydrogen bond is the strongest between the molecules of;

(A) HF

(B) HCl

(C) HBr

(D) HI

Q6. The strength of binding energy of transition elements depend upon;

(A) Number of electron pairs
(B) Number of unpaired electrons
(C) Number of neutrons
(D) Number of protons

Q7. Which set of hybrid orbitals has planar triangular shape?

(A) sp^3
(B) sp
(C) sp^2
(D) dsp^2

Q8. Preparation of vegetable ghee involves;

(A) Halogenation
(B) hydrogenation
(C) hydroxylation
(D) Dehydrogenation

Q9. Amongst the following, the compound that can be most readily sulphonated is;

(A) Toluene
(B) Benzene
(C) Nitrobenzene
(D) Chloro benzene

Q10. SN^2 reactions can be best carried out with;

(A) Primary alkyl halide
(B) Secondary alkyl halide
(C) Tertiary alkyl halide
(D) All the three

Q11. Rectified spirit contains Ethyl alcohol about;

(A) 80%
(B) 85%
(C) 90%
(D) 95%

Q12. Which of the following will have the highest boiling point?

(A) Methanal
(B) Ethanal
(C) Propanal
(D) 2-Hexanone

Q13. Which acid is used in the manufacture of synthetic fibre?

(A) Formic acid
(B) Oxalic acid
(C) Carbonic acid
(D) Acetic acid

Q14. The reaction between fat and NaOH is called;

(A) Esterification
(B) Hydrogenolysis
(C) Fermentation
(D) Saponification

Q15. Which woody raw material is used for the manufacture of paper pulp?

(A) Cotton
(B) Bagasse
(C) Poplar
(D) Rice straw

Q16. Reducing smog contains high contents of;

(A) NO_2 (B) SO_2

(C) NO (D) CO_2

Q17. Ecosystem is a smaller unit of;

(A) Lithosphere (B) Hydrosphere

(C) Atmosphere (D) Biosphere

> Note: (The End)

CHEMISTRY PAPER : PART – II (SUBJECTIVE PART)

Time Allowed: 2:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 022/1

> Note: Note:- Attempt any TWENTY TWO (22) short questions in all selecting eight from Q. 2 and Q. 3 each and six from Q. 4. (22 x 2 = 44)

> Note: SECTION – I

Q2. (Marks (2 x 8 = 16)) Write short answers of any eight questions.

(i) Why metallic character increases from top to bottom in a group?

(ii) What is Lanthanide contraction?

(iii) Give the formula of the following; (a) Spodumene (b) Magnesite

(iv) What happens when (a) Lithium carbonate is healed (b) Berylium is treated with sodium hydroxide

(v) Why aqua regia dissolve gold and platinium?

(vi) Name the allotropic forms of phosphorus.

(vii) Write the chemistry involved in Borax bead test.

(viii) How Aluminum reacts with non metals?

(ix) How chromate ions are converted into dichromate ions?

(x) What are the typical and non typical transition metals?

(xi) Write the stages involved in the manufacturing of portland cement.

(xii) Write the woody raw material used in the manufacturing of paper.

Q3. (Marks (2 x 8 = 16)) Write short answers of any eight questions.

(i) How the octane number of alkanes can be improved?

(ii) Define functional group. Give two examples.

(iii) What is iodized salt?

(iv) Define Disproportionation reaction. Give an example.

(v) Convert Ethyne into Glyoxal.

(vi) State Markownikov's rule. Give an example.

(vii) Write down four uses of Ethyne.

(viii) Define electrphile and Nucleophile.

(ix) Give the reaction of Grignard reagent with Methanal.

(x) What is saponification. Give reaction.

(xi) Define polysaccharides. Give two examples.

(xii) How polyvinyl chloride (PVC) is prepared. Give its uses.

Q4. (Marks (2 x 6 = 12)) Write short answers of any six questions.

(i) How straight chain structure was ruled out for benzene (any two points).

(ii) Convert n-Heptane to toluene.

(iii) How does ethyl alcohol react with; (a) $SoCl_2$/Pyridine (b) NH_3/THO_2.

(iv) Why and how alcohol is denatured?

(v) What is formalin? How is it prepared?

(vi) What is peptide bond? Give formula of a dipeptide.

(vii) Write down structural formula of; (a) Valeric acid (b) Acetic anhydride

(viii) Briefly discuss recycling of waste by deploymerization.

(x) What is chemical oxygen demand? How does it tell the Quality of water?

> Note: SECTION – II

> Note: Note:- Attempt any three questions. (8 x 3 = 24)

Q5. (Marks (08)) (a) Write comprehensive note on ionization energy. (04)

Q. (Marks (04)) (b) Discuss peculiar Behaviour of Boron.

Q6. (Marks (08)) (a) Describe role of lime in industries. Write only eight points. (04)

Q. (Marks (04)) (b) Write down the reactions of $KMnO_4$ with (i) H_2S (ii) $FeSO_4$ (iii) Oxalic acid (iv) KOH

Q7. (Marks (08)) (a) Define hybridization. Also explain Sp2 mode of hybridization with example of ethene. (04)

Q. (Marks (04)) (b) Using ethyl bromide as a starting material prepare following (i) Ethyl cyanide (ii) Ethene (iii) Nitroethane (iv) Ethyl acetate

Q8. (Marks (08)) (a) How addition of Halogens to alkenes takes place. Give the mechanism of reaction. (04)

Q. (Marks (04)) (b) Define Cannizzaro's reaction. Explain its mechanism with a suitable example.

Q9. (Marks (08)) (a) Write a note on sulphonation of benzene. (04)

Q. (Marks (04)) (b) Describe Lucas test to differentiate between primary, secondary and tertiary alcohols.

> Note: (The End)